DA-2

Question 1:

Code:

```cpp
#include <iostream>
#include <string>
using namespace std;

class Person {
protected:
    string name;
    int universityID;

public:
    Person(string name, int universityID) : name(name), universityID(universityID)
{}

    virtual void display() const {
        cout << "Name: " << name << endl;
        cout << "University ID: " << universityID << endl;
    }
};

class Faculty : public Person {
private:
    string rank;

public:
    Faculty(string name, int universityID, string rank) : Person(name,
universityID), rank(rank) {}

    void display() const override {
        Person::display();
        cout << "Rank: " << rank << endl;
    }
};

class UndergraduateStudent : public Person {
private:
    int yearOfStudy;

public:
    UndergraduateStudent(string name, int universityID, int yearOfStudy) :
Person(name, universityID), yearOfStudy(yearOfStudy) {}

    void display() const override {
        Person::display();
        cout << "Year of Study: " << yearOfStudy << endl;
    }
```

```
};

class GraduateStudent : public Person {
private:
    int supervisorUniversityID;

public:
    GraduateStudent(string name, int universityID, int supervisorUniversityID) :
Person(name, universityID), supervisorUniversityID(supervisorUniversityID) {}

    void display() const override {
        Person::display();
        cout << "Supervisor's University ID: " << supervisorUniversityID << endl;
    }
};

int main() {
    Faculty faculty("John Doe", 12345, "Associate Professor");
    UndergraduateStudent undergrad("Alice Smith", 54321, 2);
    GraduateStudent grad("Bob Johnson", 98765, 12345);
    cout << "Faculty Information:" << endl;
    faculty.display();

    cout << "\nUndergraduate Student Information:" << endl;
    undergrad.display();

    cout << "\nGraduate Student Information:" << endl;
    grad.display();

    return 0;
}
```

Output:

```
Name: John Doe
University ID: 12345
Rank: Associate Professor

Undergraduate Student Information:
Name: Alice Smith
University ID: 54321
Year of Study: 2

Graduate Student Information:
Name: Bob Johnson
University ID: 98765
Supervisor's University ID: 12345
harshitsingh@Harshits-MacBook-Air main %
```

Question 2:

```cpp
#include <iostream>
#include <string>
using namespace std;

class Book {
protected:
    string title;
    string author;
    int year;
public:
    Book(string title, string author, int year) : title(title), author(author),
year(year) {}

    virtual void display() {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Year: " << year << endl;
    }
};

class Reference_Book : public Book {
private:
    string topic;
public:
    Reference_Book(string title, string author, int year, string topic) :
Book(title, author, year), topic(topic) {}

    void display() override {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Year: " << year << endl;
        cout << "Topic: " << topic << endl;
    }
};

class Magazine : public Book {
private:
    string issue;
public:
    Magazine(string title, string author, int year, string issue) : Book(title,
author, year), issue(issue) {}

    void display() override {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Year: " << year << endl;
        cout << "Issue: " << issue << endl;
    }
};
```

```
int main() {
    Book book1("The Great Gatsby", "F. Scott Fitzgerald", 1925);
    Reference_Book ref_book("The C++ Programming Language", "Bjarne Stroustrup",
1985, "Programming");
    Magazine magazine("National Geographic", "Various", 1888, "April 2024");

    cout << "Book:" << endl;
    book1.display();

    cout << "\nReference Book:" << endl;
    ref_book.display();

    cout << "\nMagazine:" << endl;
    magazine.display();

    return 0;
}

            if(arr1[j]>arr1[j+1]){
                int temp=arr1[j];
                arr1[j]=arr1[j+1];
                arr1[j+1]=temp;
            }
        }
    }
    printf("Array 1:\n");
    for (int i=0;i<a;i++)
```

Output:

```
Book:
Title: The Great Gatsby
Author: F. Scott Fitzgerald
Year: 1925

Reference Book:
Title: The C++ Programming Language
Author: Bjarne Stroustrup
Year: 1985
Topic: Programming

Magazine:
Title: National Geographic
Author: Various
Year: 1888
Issue: April 2024
harshitsingh@Harshits-MacBook-Air main %
```

Question 3:

```cpp
#include <iostream>
#include <string>
using namespace std;

class ItemType {
private:
    string name;
    int deposit;
    int costPerDay;
public:
    ItemType(string name, int deposit, int costPerDay) {
        this->name = name;
        this->deposit = deposit;
        this->costPerDay = costPerDay;
    }

    // Getters
    string getName() const {
        return name;
    }

    int getDeposit() const {
        return deposit;
    }

    int getCostPerDay() const {
        return costPerDay;
    }

    // Setters
    void setName(string name) {
        this->name = name;
    }

    void setDeposit(int deposit) {
        this->deposit = deposit;
    }

    void setCostPerDay(int costPerDay) {
        this->costPerDay = costPerDay;
    }

    // Display method
    void display() const {
        cout << "Item Name: " << name << endl;
        cout << "Deposit: $" << deposit << endl;
```

```
        cout << "Cost Per Day: $" << costPerDay << endl;
    }
};

int main() {
    string name;
    int deposit, costPerDay;

    cout << "Enter item name: ";
    getline(cin, name);
    cout << "Enter deposit amount: ";
    cin >> deposit;
    cout << "Enter cost per day: ";
    cin >> costPerDay;

    ItemType item(name, deposit, costPerDay);

    cout << "\nItem Details:" << endl;
    item.display();

    return 0;
}
```

Output:

```
Enter item name: abc
Enter deposit amount: 10000
Enter cost per day: 12

Item Details:
Item Name: abc
Deposit: $10000
Cost Per Day: $12
harshitsingh@Harshits-MacBook-Air main % cd "/Users/
/Users/harshitsingh/Documents/CAT_2/main/"main
Enter item name: efg
Enter deposit amount: 12000
Enter cost per day: 13

Item Details:
Item Name: efg
Deposit: $12000
Cost Per Day: $13
harshitsingh@Harshits-MacBook-Air main %
```

Question 4:

```cpp
#include <iostream>
#include <string>
using namespace std;

class Account {
private:
    int accountID;
    string holderName;
    double balance;

public:
    // Constructor with validation
    void get(){
        cin >> accountID;
        cin >> holderName;
        cin >> balance;
    }
    Account(int accountID = -1, string holderName = "Unknown", double balance =
0.0) {
        if (accountID >= 100 && accountID <= 999) {
            this->accountID = accountID;
        } else {
            this->accountID = -1; // Invalid ID
        }
        this->holderName = holderName;
        this->balance = (balance >= 0) ? balance : 0.0;
    }

    // Method to deposit money into the account
    double credit(double amount) {
        if (amount > 0) {
            balance += amount;
        }
        return balance;
    }

    // Method to display account information
    void display() const {
        cout << "Account ID: " << accountID << endl;
        cout << "Holder Name: " << holderName << endl;
        cout << "Account Balance: $" << balance << endl;
    }
};

int main() {
    Account account;
    account.get();
```

```
    account.display();
    double depositAmount;
    cout << "\nEnter the amount: $";
    cin >> depositAmount;

    double newBalance = account.credit(depositAmount);

    cout << "New Balance: $" << newBalance << endl;

    return 0;
}
```

Output:

Question 5:

```
123
ABC
1000
Account ID: 123
Holder Name: ABC
Account Balance: $1000

Enter the amount: $4000
New Balance: $5000
harshitsingh@Harshits-MacBook-Air main % cd "/Users/
/Users/harshitsingh/Documents/CAT_2/main/"main
456
EFG
1200
Account ID: 456
Holder Name: EFG
Account Balance: $1200

Enter the amount: $1000
New Balance: $2200
harshitsingh@Harshits-MacBook-Air main %
```

```
##include <iostream>
using namespace std;

class DummyNumberValidator {
private:
    int number;
    int k;

    // Function to calculate the sum of digits of a number
    int sumOfDigits(int num) {
        int sum = 0;
        while (num > 0) {
            sum += num % 10;
            num /= 10;
        }
        return sum;
    }

public:
```

```
    // Constructor
    DummyNumberValidator(int num, int k) : number(num), k(k) {}

    // Destructor
    ~DummyNumberValidator() {
        cout << "Object destroyed." << endl;
    }

    // Method to check if the number is a valid dummy number
    bool isValidDummyNumber() {
        // Check if the number is positive
        if (number <= 0) return false;

        // Check if the sum of digits is divisible by 2, 3, or 5, and also
divisible by k
        int sum = sumOfDigits(number);
        if (sum % 2 == 0 || sum % 3 == 0 || sum % 5 == 0) {
            if (sum % k == 0) {
                return true;
            }
        }
        return false;
    }
};

int main() {
    int N, dummyNumber, k;
    cout << "Enter the value of N: ";
    cin >> N;
    cout << "Enter the dummy number: ";
    cin >> dummyNumber;
    cout << "Enter the value of k: ";
    cin >> k;

    // Create an object of DummyNumberValidator class
    DummyNumberValidator validator(dummyNumber, k);


    if (validator.isValidDummyNumber()) {
        cout << dummyNumber << " is a valid dummy number." << endl;
    } else {
        cout << dummyNumber << " is not a valid dummy number." << endl;
    }

    return 0;
}
```

Output:

```
Enter the value of N: 10
Enter the dummy number: 5
Enter the value of k: 2
5 is not a valid dummy number.
Object destroyed.
harshitsingh@Harshits-MacBook-Air main % cd "/Users/
/Users/harshitsingh/Documents/CAT_2/main/"main
Enter the value of N: 13
Enter the dummy number: 15
Enter the value of k: 4
15 is not a valid dummy number.
Object destroyed.
harshitsingh@Harshits-MacBook-Air main %
```

Question 6:

```cpp
#include <iostream>
using namespace std;

class Account {
private:
    int accountNumber;
    double amount;
    static double interestRate; // Static data member for interest rate

public:
    // Constructor
    Account(int accountNumber, double amount) {
        this->accountNumber = accountNumber;
        this->amount = amount;
    }
```

```
    void calculateAndAddInterest() {
        double monthlyInterest = (amount * interestRate) / 12.0;
        amount += monthlyInterest;
        cout << "Monthly interest added. New balance: $" << amount << endl;
    }

    static void setInterestRate(double rate) {
        interestRate = rate;
    }
};


double Account::interestRate = 0.0;

int main() {
    Account::setInterestRate(0.05);
    Account account(123456, 1000.0);
    account.calculateAndAddInterest();
    return 0;
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air main % cd "/Users
/Users/harshitsingh/Documents/CAT_2/main/"main
Monthly interest added. New balance: $1004.17
harshitsingh@Harshits-MacBook-Air main %
```

Question 7:

```
#include <iostream>
#include <string>
using namespace std;

class AutoInsurance {
```

```
private:
    int age;
    string name;
    string state;
    string insuranceID;
    static int count; // Static data member to count the number of objects

public:

    void get(){
        cin >> age;
        cin >> name;
        cin >> state;
        cin >> insuranceID;
    }
    // Default constructor
    AutoInsurance() {
        age = 0;
        count++;
    }

    // Parameterized constructor
    AutoInsurance(int age, string name, string state, string insuranceID) {
        this->age = age;
        this->name = name;
        this->state = state;
        this->insuranceID = insuranceID;
        count++;
    }

    // Destructor
    ~AutoInsurance() {
        cout << name << " is deleted" << endl;
    }

    // Function to calculate premium
    void calculatePremium() {
        if ((age >= 16 && age <= 80) && (state == "TN" || state == "KE")) {
            int basePrice = (state == "TN") ? 500 : 1000;
            int premium = basePrice + 2 * (1000 - age);
            cout << "Driver " << count << ": " << premium << endl;
        } else {
            cout << "Driver " << count << ": Invalid input" << endl;
        }
    }

    // Static member function to get the count of objects
    static int getCount() {
        return count;
    }
};
```

```cpp
// Initializing static data member
int AutoInsurance::count = 0;

int main() {

    AutoInsurance driver1;
    driver1.get();
    AutoInsurance driver2;
    driver2.get();
    AutoInsurance driver3;
    driver3.get();

    driver1.calculatePremium();
    driver2.calculatePremium();
    driver3.calculatePremium();

    // Displaying number of objects created
    cout << "No of objects: " << AutoInsurance::getCount() << endl;

    return 0;
}
```

Output:

```
25
Hari
TN
1234
40
Mary
KE
2345
15
Anu
KE
3456
Driver 3: 2450
Driver 3: 2920
Driver 3: Invalid input
No of objects: 3
Anu is deleted
Mary is deleted
Hari is deleted
harshitsingh@Harshits-MacBook-Air main %
```